Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Powerstrangers
Project / Solution Name	FutureReady MY
Name (Leader)	Mak Jia Hng
Email & Phone (Leader)	ivanjiahngmak@gmail.com (+60189708879)
University Affiliation(s)	University of Malaya, University Tunku Abdul Rahman, Xiamen University Malaysia
Live Prototype URL	https://xvu8eu1rakfbfc-8000.proxy.runpod.net
Code Repository URL (public / access granted to judges)	https://github.com/CJYJas/malaysia-workforce-platform.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

FutureReady MY uses a React 19, TypeScript, Vite, and Tailwind CSS single-page frontend with a FastAPI backend. The backend loads processed Malaysian JobStreet postings, a Coursera course catalog, and a skills taxonomy. It filters and deduplicates postings, normalises skills, calculates role-specific demand metrics, performs explainable role matching, identifies skill gaps, and retrieves traceable course recommendations. The role matchers is deterministic (70% empirical skill coverage and 30% TF-IDF similarity). The course retriever combines TF-IDF and sentence embeddings with reciprocal-rank fusion. For optional unstructured profile parsing and verified narration, the backend redacts PII before calling Qwen2.5-7B-Instruct through a RunPod Serverless vLLM endpoint. Model output is checked against the canonical vocabulary and source evidence; a deterministic fallback is used if AI is unavailable or fails validation. The backend is designed to run in a CPU RunPod pod, while GPU inference is isolated to a scale-to-zero worker.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	FastAPI backend, preprocessing matching, retrieval, and API service	PSF
FastAPI >= 0.109, Pydantic >=2.5. Uvicorn >=0.27	Typed REST API, validation, and ASGI serving	MIT
Pandas, PyArrow, PyTAML, scikit-learn	Data processing, Parquet I/o, configuration, and TF-IDF similarity	Apache 2.0
React 19, TypeScipt, Vite	Responsive web application and client-side interface	React 19, Vite: MIT TypeScript: Apache 2.0
Tailwind CSS, Radix UI, TanStack Query, Zod	Styling accessible UI components, API state, and schema validation	MIT
SQLAlchemy 2.0.41, psycopg 3.2.9, argon2-cffi 25.1	Optional account persistence PostgreSQL access, and password hashing	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless OpenAI-compatible Chat Completions API	Optional Qwen2.5-7B-Instruct skill extraction and verified narration	Sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
RunPod CPU Pod	Hosts FastAPI backend on port 8000	Sponsor-provided; CPU container; region to be selected
RunPod Serverless vLLM worker	Hosts optional Qwen2.5-7B-Instruct inference	Sponsor-provided; GPU worker; scale to zero; region to be selected
SQLite, PostgreSQL	SQLite local development; PostgreSQL-compatible account persistence when enabled	Deployment database
Replit	Fronted development workspace, artifacts, and prototype frontend deployment.	Sponsor-provided

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2.5-7B-Instruct	Qwen (deployed through RunPod vLLM)	Pre-trained; no fine-tuning recorded.	Apache 2.0
Sentence-transformers, all-MiniLM-L6-v2	Sentence Transformers, Hugging Face	Pre-trained embeddings for course retrieval; no fine-tuning recorded.	Apache 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
JobStreet all job dataset	https://www.kaggle.com/datasets/azraimohamad/jobstreet-all-job-dataset	Public Kaggle dataset; processed local copy	Public Domain / Apache 2.0	N – public job postings
Coursera dataset	https://www.kaggle.com/datasets/elvinrustam/coursera-dataset	Public Kaggle dataset; processed local copy	CC0: Public Domain	N – course catalog metadata
World Economic Forum Future of Job Report	https://www.weforum.org/publications/annual-report-2025-2026	Public report used in fronted outlook views	CC BY-NC-ND 4.0	N – published aggregate report
IT skills from jobs	https://www.kaggle.com/datasets/meerawks/it-skills-from-jobs	Public Kaggle dataset; taxonomy reference	Apache 2.0	N – no user profile data

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit AI	Frontend development assistance and prototype deployment support	Major
Anthropic Claude	Backend development support. Including code generation, debugging, and technical guidance	Major
Antigravity	Backend development support, including implementation and debugging assistance	Major
OpenAI Codex	Drafting and summarizing the PRD and Technical & Tools Declaration	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviews AI-assisted suggestion before use and remain responsible for all submitted work. Application logic is checked against the repository's data, configuration, API schemas, and tests. AI-derived skill extraction is constrained by PII redaction, canonical-vocabulary validation, verbatim evidence checks, factual-fidelity checks and deterministic fallbacks. The project does not allow an LLM to create job-market statistics course records, or unsupported claims.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Frontend development workspace and deployment	Used to develop the React/TypeScript frontend and host the prototype frontend.
RunPod	CPU pod, Serverless vLLM compute, and Network Volume	Used to deploy FastAPI, persist datasets, and optionally serve Qwen2.5-7B-Instruct inference

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and RunPod support the intended deployed demonstration. Replit provides the fronted development and deployment environment, while RunPod provides backend hosting, persistent data storage, and optional model inference. The application architecture is portable. The frontend can be hosted on another web platform, and the FastAPI service and OpenAI-compatible vLLM endpoint can be moved to another cloud provider. Therefore, sponsor resources are important for this prototype deployment.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application may receive user-entered skills, experience, education, interests, and uploaded PDF resumes. Before an optional third-party LLM request, the system redacts Malaysian NRIC numbers, phone numbers, email address, and street addresses. Profile parsing is designed to process text in memory and state that it is not used for model training. The application avoids logging raw profile text. Optional accounts use Argon2id password hashing, CSRF protection, and refresh-session rotations; SQLite is used locally and PostgreSQL-compatible persistence is available when enabled. The supplied dataset contains job postings course metadata, and skills-taxonomy records rather than candidate profiles. Data retention in a live deployment must be confirmed in its privacy notice.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

LLM extraction or narration may be unavailable, time out, return invalid JSON, introduce unsupported claims, or be affected by prompt injection. The system redacts PII, uses strict prompts, validates extracted skills against a canonical vocabulary and verbatim evidence, checks narration against source facts, and falls back to deterministic extraction or templates. Job-posting dates clustered, so the platform must not present them as smooth long-term market trends. Some roles have small samples and required warnings. Course metadata can be incomplete, and JobStreet or Coursera dataset licenses require final confirmation. AI outputs are not hiring predictions or guarantees.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [/] Any use of official SDC sponsor resources has been declared in Section 5.
- [/] The prototype described here is accessible to judges through the links provided in Section 1.
- [/] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature



Date: 24/10/2026